Year 11 Higher

REVISION PACK

Progress Check 2

ANSWERS

Units in this pack

- Graphs 5
- Handling Data 4
- Shape and Space 5

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Q	Working	Answer
adding to remove a letter		
1.1	$4x + 3y = 18$ $2x - 3y = 0$ $\times 3:\ 12x + 9y = 54$ $\times 3:\ 6x - 9y = 0$ $\text{add}:\ 18x = 54$ $x = 3$ $4 \times 3 + 3y = 18$ $y = 2$	$x = 3, y = 2$
1.2	$5x + 2y = 18$ $3x - 2y = -2$ $\times 2:\ 10x + 4y = 36$ $\times 2:\ 6x - 4y = -4$ $\text{add}:\ 16x = 32$ $x = 2$ $5 \times 2 + 2y = 18$ $y = 4$	$x = 2, y = 4$
1.3	$2x + 5y = 13$ $4x - 5y = 11$ $\times 5:\ 10x + 25y = 65$ $\times 5:\ 20x - 25y = 55$ $\text{add}:\ 30x = 120$ $x = 4$ $2 \times 4 + 5y = 13$ $y = 1$	$x = 4, y = 1$

1.4	$3x + 2y = 18$ $x - 2y = -2$ $\times 2: \ 6x + 4y = 36$ $\times 2: \ 2x - 4y = -4$ $\text{add}: \ 8x = 32$ $x = 4$ $3 \times 4 + 2y = 18$ $y = 3$	$x = 4, y = 3$
1.5	$6x + y = 11$ $2x - y = -3$ $\text{add}: \ 8x = 8$ $x = 1$ $6 \times 1 + y = 11$ $y = 5$	$x = 1, y = 5$
1.6	$5x + 4y = 6$ $3x - 4y = 10$ $\times 4: \ 20x + 16y = 24$ $\times 4: \ 12x - 16y = 40$ $\text{add}: \ 32x = 64$ $x = 2$ $5 \times 2 + 4y = 6$ $y = -1$	$x = 2, y = -1$
multiplying first		
1.7	$5x + 3y = 12$ $2x - 4y = 10$ $\times 4: \ 20x + 12y = 48$ $\times 3: \ 6x - 12y = 30$ $\text{add}: \ 26x = 78$ $x = 3$ $5 \times 3 + 3y = 12$ $y = -1$	$x = 3, y = -1$

1.8	$3x + 4y = 10$ $5x - 3y = 7$ $\times 3:\ 9x + 12y = 30$ $\times 4:\ 20x - 12y = 28$ $\text{add}:\ 29x = 58$ $x = 2$ $3 \times 2 + 4y = 10$ $y = 1$	$x = 2, y = 1$
1.9	$2x + 5y = 13$ $3x + 2y = 3$ $\times 2:\ 4x + 10y = 26$ $\times 5:\ 15x + 10y = 15$ $\text{subtract}:\ 11x = -11$ $x = -1$ $2 \times -1 + 5y = 13$ $y = 3$	$x = -1, y = 3$
1.10	$4x + 5y = 13$ $3x - 2y = 4$ $\times 2:\ 8x + 10y = 26$ $\times 5:\ 15x - 10y = 20$ $\text{add}:\ 23x = 46$ $x = 2$ $4 \times 2 + 5y = 13$ $y = 1$	$x = 2, y = 1$
1.11	$6x + 5y = 16$ $4x - 3y = -2$ $\times 3:\ 18x + 15y = 48$ $\times 5:\ 20x - 15y = -10$ $\text{add}:\ 38x = 38$ $x = 1$ $6 \times 1 + 5y = 16$ $y = 2$	$x = 1, y = 2$

1.12	$3x + 2y = 11$ $4x - 5y = 7$ $\times 5:\ 15x + 10y = 55$ $\times 2:\ 8x - 10y = 14$ $\text{add}:\ 23x = 69$ $x = 3$ $3 \times 3 + 2y = 11$ $y = 1$	$x = 3, y = 1$
problems with two unknowns		
1.13	$2n + 3p = 2.25$ $n + 4p = 1.75$ $\times 4:\ 8n + 12p = 9$ $\times 3:\ 3n + 12p = 5.25$ $\text{subtract}:\ 5n = 3.75$ $n = 0.75$ $2 \times 0.75 + 3p = 2.25$ $p = 0.25$	**notebook: 0.750 KD** **pencil: 0.250 KD**
1.14	$5a + 2m = 3.3$ $3a + 4m = 3.8$ $\times 4:\ 20a + 8m = 13.2$ $\times 2:\ 6a + 8m = 7.6$ $\text{subtract}:\ 14a = 5.6$ $a = 0.4$ $5 \times 0.4 + 2m = 3.3$ $m = 0.65$	**apple: 0.400 KD** **mango: 0.650 KD**
1.15	$3c + c = 4.1$ $2c + 5c = 4.9$ $\times 5:\ 15c + 5c = 20.5$ $\text{subtract}:\ 13c = 15.6$ $c = 1.2$ $3 \times 1.2 + c = 4.1$ $c = 0.5$	**coffee: 1.200 KD** **cake: 0.500 KD**

1.16	$2s + j = 3.5$ $s + 3j = 3$ $\times 3:\ 6s + 3j = 10.5$ $\text{subtract}:\ 5s = 7.5$ $s = 1.5$ $2 \times 1.5 + j = 3.5$ $j = 0.5$	**sandwich: 1.500 KD** **juice: 0.500 KD**
1.17	$4p + 3e = 2.8$ $2p + 5e = 3.5$ $\times 5:\ 20p + 15e = 14$ $\times 3:\ 6p + 15e = 10.5$ $\text{subtract}:\ 14p = 3.5$ $p = 0.25$ $4 \times 0.25 + 3e = 2.8$ $e = 0.6$	**pencil: 0.250 KD** **eraser: 0.600 KD**
1.18	$3b + 2m = 10$ $b + 2m = 5$ $\times 2:\ 6b + 4m = 20$ $\times 2:\ 2b + 4m = 10$ $\text{subtract}:\ 4b = 10$ $b = 2.5$ $3 \times 2.5 + 2m = 10$ $m = 1.25$	**book: 2.500 KD** **magazine: 1.250 KD**
adding the equations		
1.19	$3x + y = 7$ $2x - y = 3$ $\text{add}:\ 5x = 10$ $x = 2$ $3 \times 2 + y = 7$ $y = 1$	$x = 2, y = 1$

1.20	$4x + y = 11$ $x - y = -1$ add : $5x = 10$ $x = 2$ $4 \times 2 + y = 11$ $y = 3$	$x = 2, y = 3$
1.21	$2x + y = 8$ $5x - y = 13$ add : $7x = 21$ $x = 3$ $2 \times 3 + y = 8$ $y = 2$	$x = 3, y = 2$
1.22	$x + y = 5$ $3x - y = 11$ add : $4x = 16$ $x = 4$ $1 \times 4 + y = 5$ $y = 1$	$x = 4, y = 1$
1.23	$5x + y = 9$ $2x - y = -2$ add : $7x = 7$ $x = 1$ $5 \times 1 + y = 9$ $y = 4$	$x = 1, y = 4$
1.24	$3x + y = 11$ $4x - y = 3$ add : $7x = 14$ $x = 2$ $3 \times 2 + y = 11$ $y = 5$	$x = 2, y = 5$

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Q	Working	Answer
questionnaires		

2.1		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
2.2		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 7; no box for 0**
2.3		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6 KD; no box for 0**
2.4		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
2.5		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 30 KD; no box for 0**
2.6		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 4; no box for 0**
averages from a frequency table		
2.7	**(a)** $\text{highest frequency} = 8$ **(b)** $0 \times 3 + 1 \times 7 + 2 \times 8 + 3 \times 5 + 4 \times 2 = 46$ $46 \div 25 = 1.84$	**(a) 2** **(b) 1.84**
2.8	**(a)** $\text{highest frequency} = 9$ **(b)** $1 \times 2 + 2 \times 9 + 3 \times 6 + 4 \times 4 + 5 \times 4 = 74$ $74 \div 25 = 2.96$	**(a) 2** **(b) 2.96**
2.9	**(a)** $\text{highest frequency} = 8$ **(b)** $0 \times 6 + 1 \times 5 + 2 \times 4 + 3 \times 8 + 4 \times 2 = 45$ $45 \div 25 = 1.8$	**(a) 3** **(b) 1.8**
2.10	**(a)** $\text{highest frequency} = 9$ **(b)** $0 \times 5 + 1 \times 9 + 2 \times 4 + 3 \times 1 + 4 \times 1 = 24$ $24 \div 20 = 1.2$	**(a) 1** **(b) 1.2**

2.11	**(a)** highest frequency $= 8$ **(b)** $0 \times 2 + 1 \times 4 + 2 \times 8 + 3 \times 4 + 4 \times 2 = 40$ $40 \div 20 = 2$	**(a) 2** **(b) 2**
2.12	**(a)** highest frequency $= 8$ **(b)** $0 \times 3 + 1 \times 8 + 2 \times 6 + 3 \times 2 + 4 \times 1 = 30$ $30 \div 20 = 1.5$	**(a) 1** **(b) 1.5**
a missing number from the mean		
2.13	$5 \times 16 = 80$ $12 + 14 + 19 + 20 = 65$ $80 - 65 = 15$	**15**
2.14	$6 \times 11 = 66$ $8 + 9 + 13 + 14 + 10 = 54$ $66 - 54 = 12$	**12**
2.15	$5 \times 23 = 115$ $18 + 25 + 27 + 20 = 90$ $115 - 90 = 25$	**25**
2.16	$5 \times 12 = 60$ $9 + 11 + 14 + 15 = 49$ $60 - 49 = 11$	**11**
2.17	$4 \times 20 = 80$ $18 + 25 + 16 = 59$ $80 - 59 = 21$	**21**
2.18	$6 \times 15 = 90$ $12 + 18 + 14 + 20 + 11 = 75$ $90 - 75 = 15$	**15**

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Q	Working	Answer
the angle in a semicircle		

3.1	angle $ACB = 90^\circ$ $z = 180 - 90 - 31$ $z = 59^\circ$	$z = 59°$**. The angle in a semicircle is 90°.**
3.2	angle $ACB = 90^\circ$ $z = 180 - 90 - 42$ $z = 48^\circ$	$z = 48°$**. The angle in a semicircle is 90°.**
3.3	angle $ACB = 90^\circ$ $z = 180 - 90 - 23$ $z = 67^\circ$	$z = 67°$**. The angle in a semicircle is 90°.**
3.4	angle $ACB = 90^\circ$ $z = 180 - 90 - 36$ $z = 54^\circ$	$z = 54°$**. The angle in a semicircle is 90°.**
3.5	angle $ACB = 90^\circ$ $z = 180 - 90 - 19$ $z = 71^\circ$	$z = 71°$**. The angle in a semicircle is 90°.**
3.6	angle $ACB = 90^\circ$ $z = 180 - 90 - 48$ $z = 42^\circ$	$z = 42°$**. The angle in a semicircle is 90°.**
angles in the same segment		
3.7	$w = 46^\circ$	$w = 46°$**. Angles in the same segment are equal.**
3.8	$w = 29^\circ$	$w = 29°$**. Angles in the same segment are equal.**
3.9	$w = 53^\circ$	$w = 53°$**. Angles in the same segment are equal.**
3.10	$w = 35^\circ$	$w = 35°$**. Angles in the same segment are equal.**
3.11	$w = 41^\circ$	$w = 41°$**. Angles in the same segment are equal.**
3.12	$w = 62^\circ$	$w = 62°$**. Angles in the same segment are equal.**
cyclic quadrilaterals		

3.13	$x = 180 - 124$ $x = 56^\circ$	$x = 56°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
3.14	$x = 180 - 103$ $x = 77^\circ$	$x = 77°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
3.15	$x = 180 - 97$ $x = 83^\circ$	$x = 83°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
3.16	$x = 180 - 109$ $x = 71^\circ$	$x = 71°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
3.17	$x = 180 - 131$ $x = 49^\circ$	$x = 49°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
3.18	$x = 180 - 86$ $x = 94^\circ$	$x = 94°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
the alternate segment theorem		
3.19	$y = 68^\circ$	$y = 68°$**. Alternate segment theorem.**
3.20	$y = 57^\circ$	$y = 57°$**. Alternate segment theorem.**
3.21	$y = 74^\circ$	$y = 74°$**. Alternate segment theorem.**
3.22	$y = 63^\circ$	$y = 63°$**. Alternate segment theorem.**
3.23	$y = 49^\circ$	$y = 49°$**. Alternate segment theorem.**
3.24	$y = 78^\circ$	$y = 78°$**. Alternate segment theorem.**
the angle at the centre		
3.25	$150 \div 2 = 75^\circ$	**75°. The angle at the centre is twice the angle at the circumference.**
3.26	$96 \div 2 = 48^\circ$	**48°. The angle at the centre is twice the angle at the circumference.**

3.27	$124 \div 2 = 62^\circ$	**62°. The angle at the centre is twice the angle at the circumference.**
3.28	$108 \div 2 = 54^\circ$	**54°. The angle at the centre is twice the angle at the circumference.**
3.29	$136 \div 2 = 68^\circ$	**68°. The angle at the centre is twice the angle at the circumference.**
3.30	$84 \div 2 = 42^\circ$	**42°. The angle at the centre is twice the angle at the circumference.**